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A Reflective Journey: Navigating Your Cumulative Experience at Iowa State University

Through my time at Iowa State I have evolved a lot as a person and as an engineer. When I started my degree as a computer engineer, I focused mainly on learning technical concepts and completing assignments correctly. Over time, however, I got involved with research, internship, tutoring, projects, student organizations and a lot more. These experiences showed me that engineering, along with individual learning, also requires communication, judgement and how a solution to one thing affects another. My biggest area of growth has been learning how to approach a given problem where no step by step answer is given to it. Which I do by defining the issue, find reliable information, test a solution and then revise it and repeat it if necessary.

Applying my classroom learning to engineering problems is part of my daily routine now. I will list examples of these as follows. In a class I am currently taking which has a lab component to it, I implemented parts of a convolutional neural network in C++ and examined how each layer affects the actual inference time involved. The engineering problem lies in the fact that I had to connect the mathematical operation of convolution to actual code that handled inputs, weights, biases and so on rather than just use a simple library call to get it done. I also had to profile the model which showed me that optimization requirements will come through in the future as a need. At the same time, in our senior design project we are working with Buildertrend to generate AI renderings from pans from 2d visualizations. The engineering

challenge here is to determine what the input plan contains and get a useful rendering that determines what the input plan contains, and gives out a well enough template that a potential client would like. While these are ongoing projects, I have come to appreciate that solving problems involves both building a system and defining what success looks like in that system, as working without a goal takes one no where.

I have also taken the time to learn independently from outside sources. In my ARA research I worked with tools that were unfamiliar to me including Docker, Ericsson Moshell environments, and network infrastructure wherein these were things I learned from scratch. When an implementation did not work as expected, I inspected error messages, consulted documentation, logically described the causes and the potential solutions and then tested on these. And this happened a lot when I built something, it didn't work as expected on the first pass. These experiences along with guidance from professors, mentors and peers, showed me that engineers combine written sources, expert guidance and own tests to make things work and pass through. In addition to this, my current research and a Boeing Research fellow, these same concepts are being reinforced in real time.

Piggybacking off this, communication and teamwork were also something that came in hand in hand. Giving examples to these, while writing ARA technical documentation, it required me to think about the person who would be reading these and using these. So putting myself in their shoes, I try to direct this and other works where I make documentation such that it is straightforward for a person to understand. Adding to this, as a LEAD peer tutor, I have come to learn that a simple explanation works on even the most complex topics. With that being said, what may work for one student may not work for another, so bridging that gap and learning the

best ways to communicate a solution or give out help is one thing I have learned and become much better at it, and I am sure this will be of great benefit to me in the future as a communicator. Other experiences where I have come to appreciate teamwork and communication are my internship at Corteva, group projects and more. These experiences helped me appreciate communication as a subset of teamwork and its immense importance in an engineering setting. Poor communication can lead to errors even if everyone is technically sound.

These experiences also got me thinking about my ethical responsibility as an engineer. In wireless networking reliability matters as failures affect people trying to communicate or use the services. That would give engineers the responsibility to test changes carefully and understand the effects of a configuration. With respect to senior design, it is my responsibility as an engineer to give a rendered plan that generates confidence in details that may have been missed. The system that generates these plans does not do so in a way such that it takes creative liberties and makes something completely unrelated. All of this ties back to the fact that technically impressive results are not enough, an engineer must connect technical accuracy with the potential societal impact i.e. the people who are affected by this system, the economic impact and the impact as well. While my experiences have been small and not vast enough where I can say that anything I made has changed the environment. In general things engineers build can affect the environment, and it is our duty to make sure that we adhere to it in a way that we can make sure we are not damaging it.

While I have covered my professional experiences in the above paragraph, I want to talk about my leadership experiences through my experiences with the Indian Student association. Here I have been serving as the treasurer for more than a year. Here I track expenses of the club, manage a budget, communicate decisions to other members and so on and so forth. This responsibility taught me the importance of accuracy, accountability, communication and more. In terms of engineering, it connects in the sense that projects also have limited resources so teams must make it work with limited resources. At the end of this opportunity I appreciate a lot more about contributing reliable work behind the scenes and making sure the entire team succeeds.

In conclusion, my college experience has changed how I define engineering ability. Technical knowledge matters but also learning independently, communicating clearly and considering the effects of the work done. With this said, I still need work on improving turning open ended goals into testable requirements and explaining technical decisions to people with different backgrounds. As I complete senior design and move toward graduate studies, I plan on building more and refining these skills and overall become a better engineer.